White Paper

STORAGE QUOTA MANAGEMENT AND PROVISIONING WITH EMC ISILON SMARTQUOTAS

Abstract

This white paper describes how EMC Isilon SmartQuotas software helps enterprises drive down storage costs with a simple to use, highly scaleable and flexible storage quota and provisioning application for scale-out storage environments.

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Introduction

File-based data continues to grow at a remarkable rate—nearly doubling each year—making the need for optimized data storage and its management more important than ever.

To help enterprises maximize the long-term value of their critical business data and drive down storage management cost and complexity, EMC Isilon offers SmartQuotas: a simple, scalable and flexible quota management and provisioning software application that integrates with the EMC Isilon OneFS® operating system. SmartQuotas allows administrators to **control and limit storage usage** across their organization and provision a single pool of Isilon clustered storage to best meet their unique storage challenges. SmartQuotas also provides **Thin Provisioning** to present more storage capacity to applications and users than is physically present. SmartQuotas requires a separate license to be used on a cluster node.

This paper examines how SmartQuotas can be applied by storage administrators to establish storage usage quotas and how these quotas can be enforced. The paper also describes notification and reporting options available to streamline management of scale-out NAS storage environments.

Overview of SmartQuotas

A quota by definition is the permissible share or proportional part of a total. Applied to storage requirements, it is the amount of storage capacity that is permissible to a certain entity within the Isilon cluster. At its core, a quota system is a combination of accounting, enforcement and reporting. **Accounting** refers to keeping counts of blocks, inodes, and files owned by resource entities like users, groups and directories. **Enforcement** refers to setting and forcing limits for certain counts and **Reporting** refers to the mechanism by which such enforcement can be conveyed to the administrators or users.

To apply SmartQuotas, two types of capacity quotas need to be considered: **Accounting Quotas** and **Enforcement Quotas**:

Accounting Quotas

Accounting quotas monitor but do not limit disk storage utilization, are useful for auditing, planning, or billing purposes. For example, with SmartQuotas accounting quotas, you can:

- Track the amount of disk space used by various users or groups to bill each entity for only the disk space used.
- Review and analyze reports that help you identify storage usage patterns. In turn, these can used to define storage policies for the organization and educate end users of the file system about using storage more efficiently.
- Intelligently plan for capacity expansions and future storage needs.

Enforcement Quotas

There are four levels of enforcement quotas: hard, soft, advisory or none. Each of these are described in Table 1.

Enforcement Level	Description
Hard	A limit that cannot be exceeded. If an operation, such as a file write, causes a quota target to exceed a hard quota, the operation fails, an alert is logged to the cluster, and a notification is issued to any specified recipients.
Soft	A limit that can be exceeded until a grace period has expired. When a soft quota is exceeded, an alert is logged to the cluster and a notification is issued to any specified recipients; however, data writes are permitted during the grace period. If the soft threshold is still exceeded when the grace period expires, data writes fail, and a hard-limit notification is issued to any specified recipients.
Advisory	An informational limit that can be exceeded. When an advisory quota threshold is exceeded, an alert is logged to the cluster and a notification is issued to any specified recipients. Reaching an advisory quota threshold does not prevent data writes.
None	No enforcement; It is an accounting only quota

Table 1: Quota Enforcement Levels

How does SmartQuotas Accounting work?

Domains

SmartQuotas works on the basis of **domains**, which is the cornerstone of the quota accounting. A domain is a set of resources and an accounting of each resource type for that set. SmartQuotas use the concept of a domain to represent all traditional quota types, but also gives the Administrator, ways to express complex domains in a straightforward manner. A simple domain defined as “identity@directory” could be the set of files under “directory”, owned by “identity”, which could be a user or group. The files accounted include all files reachable from the given path without following soft links. “identity” can be ALL (and /ifs is an effective ALL for directory).

SmartQuotas can be applied to a variety of entities as shown in Table 2.

Entity/Resource Type	Description
Directory	A specific directory and all its subdirectories.
User	A specific user.
Group	All members of a specific group.

Table 2: Entity and Resource Types

With SmartQuotas it is possible to form more traditional domain types fairly quickly by using ALL. The following are examples of domain types:

- All files belonging to user John: user:John@/ifs
- All files under /ifs/cs, belonging to any user: ALL@/ifs/cs.
- All files under /ifs/cs that belong to user John: user:John@/ifs/cs

Domains cannot be formed on anything but directories. More specifically, domains are associated with directories, not directory paths. If the domain is ALL@/ifs/cs/data, but /ifs/cs/data gets renamed to /ifs/cs/files, the domain stays with the directory.

Domains can be nested and may overlap. For example:

- A hard quota is set on /ifs/data/finance for 5TB
- 1 TB soft quotas are then placed on individual users in the finance department
- This ensures that the finance directory as a whole never exceeds 5TB while limiting the users in the finance department to 1TB each.

Default Domains and Linked Quotas:

A default domain is a domain that does not account for any specific set of files, but instead specifies policy for new domains that match a specific trigger. In short, default domains are templates for actual domains. SmartQuotas use the identity notation default-user and default-group to describe domains with default user and group policies. The domain default-user@/ifs/cs becomes specific-user@/ifs/cs for each specific-user that is not otherwise defined. All enforcements on default-user are copied to specific-user when specific-user allocates within the domain, and the new inherited domain quota is termed as a Linked Quota. There may be overlapping defaults (i.e. default-user@/ifs and default-user@/ifs/cs may both be defined).

If enforcements on defaults change, SmartQuotas will propagate the changes to the Linked Quota domains asynchronously with some lag before updates take effect. If a default domain is deleted, SmartQuotas will delete all children marked as inherited. As an option, an administrator may delete the default without deleting the children, but this will break inheritance on all inherited children.

Accounting

Domains account for physical space and logical space resources:

- **Physical Space** - All space used to store files and directories. This includes both data and metadata, but only those pieces within the domain in question. This space does not account for sparse regions. For example, only the data portions of a file will be accounted, snapshots of files and directories will not be accounted.
- **Logical Space** - The sum of all file sizes. This does not include file metadata and sparse regions..

Logical space is meant to account for space occupied by user data, and not for the overhead of keeping that space safe. The impetus for logical space is that normal file systems have traditionally provided quota systems that are physical, but, because they lack the same protection mechanisms, the typical accounting is fairly close to the user data plus some metadata overhead. However, typical quota systems still charge accounts for things like the metatree. Because SmartQuotas offer both logical and physical, it goes a step further with logical space and is used to account only the logical space, the data within files. SmartQuotas cannot, however, be used to charge for user data alone. This would open up cases where an adversarial user could hide away data in strange places. For example, if directory entries were not accounted for in some fashion, data could be hidden in directories using zero length files. To avoid this, SmartQuotas charges for certain metadata attributes, as defined in Table 3.

Data Type	Accounting
Regular file data	Every non-sparse 8k region of the file
Directories	Sum of all directory entries
Symlinks	Data Size
ACL and similar	Data Size
ADS	Each ADS is charged as a file and a container as a directory

Table 3: Data accounting for different entity types

Here are some additional interesting cases during Quota accounting:

Hard Links - Each logical inode is accounted exactly once in every domain to which it belongs. If an inodeis present in multiple domains, it is accounted in multiple domains. Alternatives such as shared accounting were considered. However, if inodes are not accounted once in every domain, it is possible for the deletion of a hard link in one domain to put another domain over quota.

Alternate Data Stream (ADS) - A file with ADS is accounted as the sum of the resource usage of the individual file, the usage for the container directory, and the usage for each ADS. SmartQuotas handles the rename of a file with ADS synchronously despite the fact that the ADS container is just a directory. SmartQuotas will store an accounting summary on the ADS container to handle renames.

Directory Rename – A directory rename presents a unique challenge to a per-directory quota system. Renames of directories within a domain are trivial - if both the source and target directories have the same domain membership, no accounting changes. However, non-empty directories are not permitted to be moved when the SmartQuotas configuration is different on the source and the target parent directories. If a user trusts the client operating systems to copy files and preserve all the necessary attributes, then the user may set dir_rename_errno to EXDEV, which causes most Unix and Windows clients to do a copy and delete of the directory tree to effect the move.

Snapshot Accounting – Quota Domains can include snapshot usage in its accounting. SmartQuotas will only support snapshots created after the quota domain was created. This is because determining quota governance (including QuotaScan job) for existing snapshots is a very time and resource consuming operation. As most administrators cycle their snapshots through timed expirations, this means that SmartQuotas will eventually accrue enough accounting information to include the entire set of relevant snapshots on the system.

QuotaScan Job

The QuotaScan is the cluster maintenance process responsible for scanning the cluster to performing accounting activities to bring the desired governance to each inode. In essence it is a distributed tree walk that is performed based on the state of the domain.

A domain may be in one of three **accounting states**:

- **Ready** - A domain in the ready state is fully accounted. SmartQuotas displays "ready" domains in all interfaces, and all enforcements apply to such domains.
- **Accounting** - Any time a domain's accounting must be updated by a QuotaScan, the domain is placed in the Accounting state. SmartQuotas displays accounting domains in all interfaces, including usage data, but indicate they are in the process of being "Accounted". SmartQuotas applies all enforcements to accounting domains, even when it might reject an allocation that would have proceeded if it had completed the QuotaScan.
- **Deleting** - Domain removal may be a lengthy process. After a request to delete a domain, SmartQuotas will place the domain in the deleting state until tear-down is complete. SmartQuotas hide domains in the deleting state from all interfaces, and the top-level directory of a domain may be deleted while the domain is still in the deleting state (assuming there are no domains in "Ready" or "Accounting" state defined on the directory). No enforcements are applied for domains in "Deleting" state.

A Quotascan is performed when the domain is in an **Accounting State**. This can occur during **quota creation** to account the new domain if a quotas has been set for the domain and **quota deletion** to un-account the domain. A QuotaScan is required when creating a quota on a non-empty directory. If quotas are created up-front on an empty directory, no QuotaScan is necessary.

In addition, a QuotaScan job may be started from the command line (using the "isi job" command). Any path specified on the command line is treated as the root of a tree that should be processed. This is provided primarily as a means to re-scan a directory or maintenance reasons.

SmartQuotas Enforcement

Now that accounting is relatively well defined, quotas are simply a matter of enforcement on accounting. Note that since all domains account for every resource type, a domain may be created for accounting only, with no enforcement.

Enforcement Types & States

Advisory - When an advisory threshold is reached, a notification condition is met. Notification rules are described in the next section. Advisory thresholds are parameterized by threshold value. There can only be one advisory threshold per domain.

- **Soft** - When a soft threshold is reached, a notification is sent as in the case of advisory thresholds. In addition, if the usage stays over the soft threshold past a specified grace period, the soft threshold becomes a hard threshold. If the usage goes below the soft threshold, the timer for the grace period is reset. Soft thresholds are parameterized by threshold value and grace period. There can only be one soft threshold per domain. The grace period cannot be infinite, nor can it be zero. For those cases, advisory and hard thresholds are available, respectively.
- **Hard** - When a hard threshold is reached, a notification is sent as for advisory and soft thresholds. In addition, all further allocations within the domain fail. Hard thresholds are parameterized by threshold value. There can only be one hard threshold per domain.

SmartQuotas defines the following enforcement states:

- **U** (Under) - If the usage is less than the enforcement threshold, the enforcement is in state U.
- **O** (Over) - If the usage is greater than the enforcement threshold, the enforcement is in state O.
- **E** (Expired) - If the usage is greater than the soft threshold, and the usage has remained over the enforcement threshold past the grace period expiration, the soft threshold is in state E. If an administrator modifies the soft threshold but not the grace period, and the usage still exceeds the threshold, the enforcement is in state E.

Exceptions to enforcement

There are a few exceptions to enforcement of Quotas including the following scenarios:

- If a domain has an accounting only quota, enforcements for the domain are not applied.
- Any administrator action may push a domain over quota. Examples include changing protection, taking a snapshot, removing a snapshot, etc. The administrator may write into any domain without obeying enforcements.

- Any system action may push a domain over quota, including repair etc. OneFS maintenance processes are as powerful as the administrator.

Quota Notification

A crucial part of the quota system is to provide user notifications regarding enforcement violations, both when a violation event occurs and while violation state persist on a scheduled basis. An enforcement may have several notification rules associated with it. Each notification rule specifies a condition and an action to be executed when the condition is met. Notification rules are considered part of enforcements. Clearing an enforcement also clears any notification rules associated with it.

Notification settings

Enforcement quotas support the notification settings described in Table 4. Note that any given quota can use only one of these settings.

Notification Setting	Description
Global default	Uses the global default notification for the specified type of quota.
None	Disables all notifications for the quota.
Custom – Basic	Enables administrators to create basic custom notifications that will apply to a specific quota. Basic notifications can be configured for any or all of the threshold types (hard, soft, or advisory) for the specified quota.
Custom - Advanced	Enables the creation of advanced custom notifications that will apply to a specific quota. Advanced notifications can be configured for any or all of the threshold types (hard, soft, or advisory) for the specified quota.

Table 4: Notification Settings

Notification conditions

A notification condition is an event that can trigger an action specified in a notification rule. A rule may specify a schedule (e.g. “every day at 1:00 AM") for executing an action or immediate notification of certain classes of state transitions. Examples of notification conditions include:

- Quota exceeded (when enforcement states is “O” – Over)
- Grace period specified for a Soft Advisory limit has expired.
- Write Access denied when attempting to write to file after quota has exceeded the limit.

Notification Actions

Each notification rule is allowed to execute one or none of each of the Notification Actions shown in Table 5.

Notification Action	Description
Alert	Send an alert for one of the actions described in the next section
Email Manual Address	Email to a specific address
Email Owner	Email owner mapping based on identity source

Table 5: Notification Actions

The email owner mapping is as follows:

- **Active Directory**: Lookup is performed against the domain controller (DC). If the user does not have an email setting, a configurable transformation from user name and DC fully qualified domain name is performed to generate an email address .
- **LDAP**: LDAP user email resolution is similar to AD users. In this case, only the email attribute looked up in the LDAP server is configurable by an administrator based on the LDAP schema for the user account information.
- **NIS**: Only the configured email transformation for the NIS fully qualified domain name is used.
- **Local users**: Only the configured email transformation is used.

How does Notification work?

The Quota notification is handled by a notification daemon which performs the following functions:

- Process Kernel notification events that get sent out. They are matched to notification rules to generate instant notifications (or other actions as specified in the notification rule)
- Process notification schedules – The daemon will check notification rules on a scheduled basis. These rules specify what violation condition should trigger a notification on a regular scheduled basis.
- Execute notifications based on rule configuration to generate emails or alert notifications
- Manage persistent notification states by recording the state persistently so that in the event of a restart, outstanding events that have yet to be notified, are handled.
- Process rescan requests by responding to kernel requests for Quota Rescan (when quotas are created or modified)

System Alerts

Various system alerts are sent out to the standard cluster Alerting system when specific events occur. These are described in Table 6.

Name	Level	Event Description
NotifyFailed	Warning	An attempt to process a notification rule failed externally, such as an undeliverable e-mail.
NotifyConfig	Warning	An attempt to process a notification rule failed due to configuration, such as nonexistent user, or missing e-mail address.
ThresholdViolation	Info	A quota threshold was exceeded. The conditions under which this alert is triggered are defined by notification rules.
DomainError	Error	An invariant was violated that resulted in a forced domain rescan.

Table 6: System Alerts

Quota Reports

SmartQuotas software supports flexible reporting options that enable administrators to more effectively manage cluster resources and analyze usage statistics. The goal of Quota Reporting is to provide a summarized view of the past or present state of the Quota Domains. There are three methods of data collection and reporting that are supported:

- **Scheduled reports** are generated and saved on a regular interval.
- **Ad-hoc reports** are generated and saved per request of the user.
- **Live reports** are generated for immediate and temporary viewing.

The CLI export functionality makes use of the same data generation and storage format as quota reporting, but should not require any extra requirements beyond the three types of reports. After the collection of the raw reporting data, data summaries can be produced given a set of filtering parameters and sorting type.

Reports can be viewed from historical sampled data or a live system. In either case, the reports are views of usage data at a given time. SmartQuotas does not provide reports on aggregated data over time (i.e. trending reports). However, the raw data can be used by a Quota Administrator to answer trending questions.

Quota report format

The storage format needs to handle consistency within domains and usability outside the context of a changing filesystem. The quota report format is a serialization of the quota configuration. A quota report is a time-stamped XML file that starts off with global configuration settings and global notification rules:

```
<quota-report time="1161852536">
<global-config>INSERT QUOTA GLOBAL CONFIG</global-config>
<domains>
...
</domains>
</quota-report>
```

When listing domains, a few special considerations are made to handle LIN-to-path mapping, UID/GID-to-name mapping, notification rules and inherited domains. Both inode/path as well as ID/name are stored with each domain. The path and name are resolved at the time of writing the domain to the report. This mitigates problems that arise from changes to the mappings during or after domain iteration. Quota Notification Rules are read and inserted into a domain entry only if the domain is not inherited. This method mitigates changes to Quota Notification Rules by accessing each rule set only once as well as limiting performance impacts of reading the Quota Notification Rules with each and every domain.

Quota Report Management

Report management needs to provide proper locking during report generation, atomic publishing of a completed reports and deletion of old reports. This is to ensure that the report is generated as an unambiguous point-in-time report of the quota/data usage. Scheduled and ad-hoc reports both maintain a set of completed reports that are rotated out as new reports become available. A set is parameterized by path, filename prefix and retained count. The temporal ordering is determined through a timestamp in the filenames. A single lock file per set is used to serialize access. Though only addition to the set needs to be serialized, creation of reports is serialized under the same lock to simplify the process.

The following are the steps followed given a path of /ifs/reports, filename prefix of ***schedule_quota_report*** and a retention count of 3:

1. Acquire a lock on /ifs/reports/.scheduled_quota_report.lock
2. Open /ifs/reports/.scheduled_quota_report.tmp truncated
3. Execute the API function to generate the quota report on (quota_report_generate) on the temporary file
4. Close the temporary file and record the time
5. Move the temporary file to /ifs/reports/scheduled_quota_report_‹timestamp›.xml
6. Perform pruning to keep to the retention count of 3
7. Release lock

Use Cases for SmartQuotas

Use Case 1: Quota management - A university wants to give their students and groups a fixed amount of storage to control and keep storage growth in check

- The storage administrator wants to know how much each student is consuming & limit them
- Set default user hard or soft quotas
- Email alerts to students to encourage self-cleanup of file usage

- **Use Case 2: Thin Provision and Scale-Out** – A company in the Media and Entertainment industry wants to overprovision storage and only add capacity when needed.
  - The company buys a 20TB cluster and gives 100 users 500GB each ~ effectively thin provisioned 50TB's
  - Set directory quotas for each user with hard quota of 500GB each
  - Set up system alert to the storage admin to add capacity (nodes) when the 20TB is 3/4 full
  - Scale-out effectively by adding additional capacity only when needed.

Summary

Traditional systems with volume quotas are limited to a single storage device— and to a single volume. Volume-level solutions are limited in management flexibility once they are initially set up. For example, typical implementations require the management of quotas across different volumes, multiple storage systems and multiple file systems. This approach creates challenges when making changes to resources, tracking specific users and groups, moving directory structures or moving data between file systems and volumes. As more storage silos are added, the complexity only increases.

With EMC Isilon SmartQuotas, these management challenges are eliminated. Because SmartQuotas is fully cluster-aware and spans across all nodes, regardless of the cluster size, IT managers can administer quotas from a single point of view, provision storage exactly how they want, and change quota policy options on demand.

Administrators can define named quotas for specific individual users or groups, or create default quotas that control disk usage for anyone accessing the cluster. Administrators can manage storage across their enterprise so that specific users and groups are only allowed to "see" the storage they have been provisioned. Hard, soft and advisory limits can be set across the organization for specific users and groups, and across the various directory structures.

Administrators can also configure alerts and send e-mail notifications to end-users letting them know that quota limits are approaching, enforcing hard stops on writes or providing a grace period of several days before enforcing thresholds. When new users need to be added or directory structures need to be changed, SmartQuotas allows IT managers to change their quota policies on the fly—easily meeting the changing storage demands of the enterprise.